

Revision del dataset Iris paso a paso

Cada seccion deja una tabla en el workspace: T0_crudo, T1_outliers, T2_balanceado, T3_estandarizado y T4_normalizado. R3 entrena el perceptron y el Adaline sobre cada una.

```
clear; close all; clc
```

Paso 0: dato crudo

```
iris = load('iris.dat');
T = array2table(iris(:,1:4), 'VariableNames',
{'largo_sepalo', 'ancho_sepalo', 'largo_petal', 'ancho_petal'});
T.clase = categorical(iris(:,5), 1:3, {'setosa', 'versicolor', 'virginica'});
vars = T.Properties.VariableNames(1:4);
T0_crudo = T;
summary(T0_crudo)
```

T0_crudo: 150x5 table

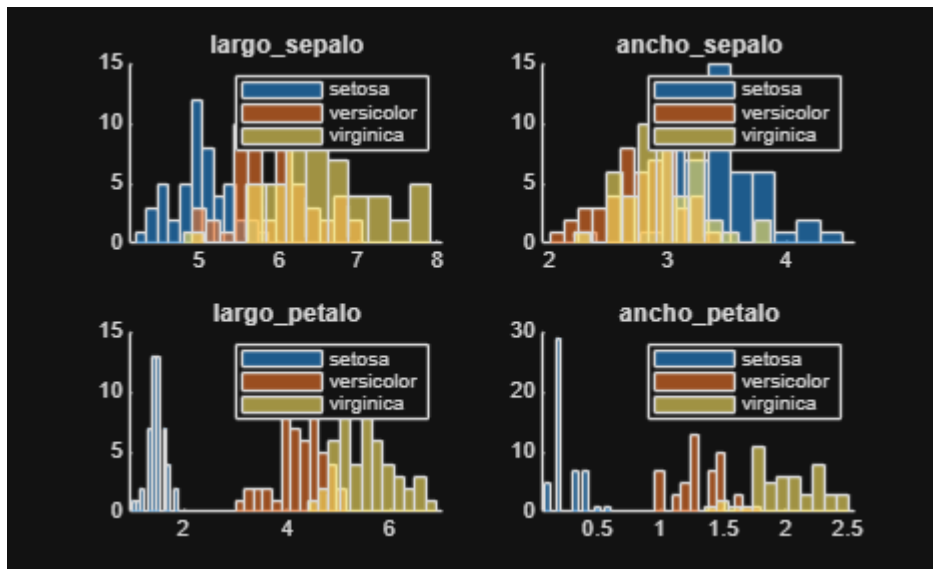
Variables:

```
largo_sepalo: double
ancho_sepalo: double
largo_petal: double
ancho_petal: double
clase: categorical (3 categories)
```

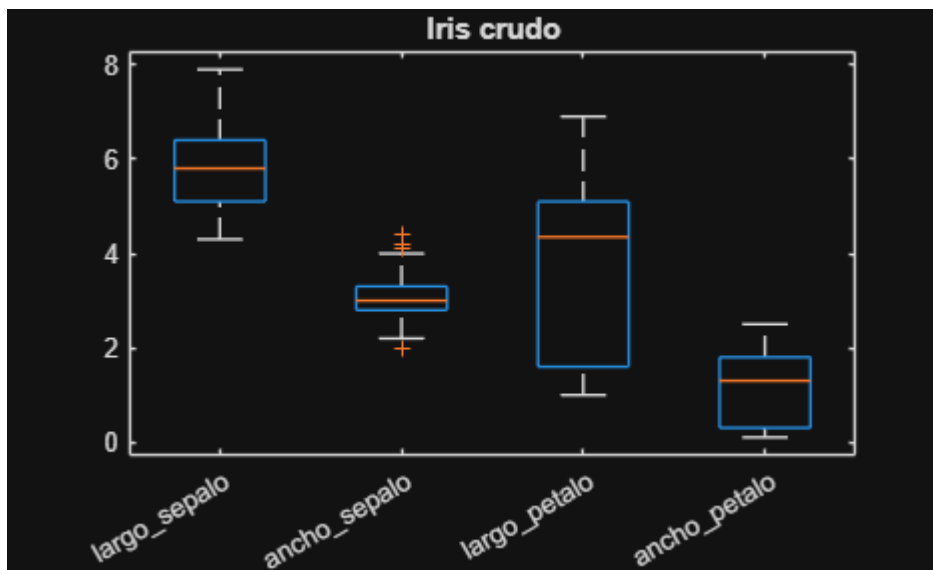
Statistics for applicable variables:

	NumMissing	Min	Median	Max	Mean	Std
largo_sepalo	0	4.3000	5.8000	7.9000	5.8433	0.8281
ancho_sepalo	0	2	3	4.4000	3.0573	0.4359
largo_petal	0	1	4.3500	6.9000	3.7580	1.7653
ancho_petal	0	0.1000	1.3000	2.5000	1.1993	0.7622
clase	0					

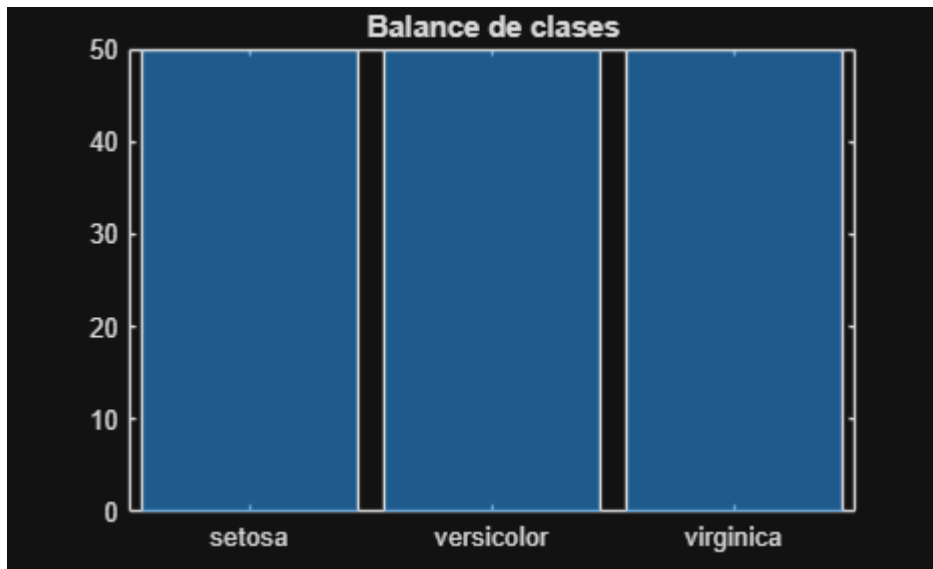
```
figure
for k = 1:4
    subplot(2,2,k); hold on
    for c = categories(T.clase)'
        histogram(T.(vars{k})(T.clase == c{1}), 12, 'DisplayName', c{1})
    end
    title(vars{k}, 'Interpreter', 'none'); legend
end
```



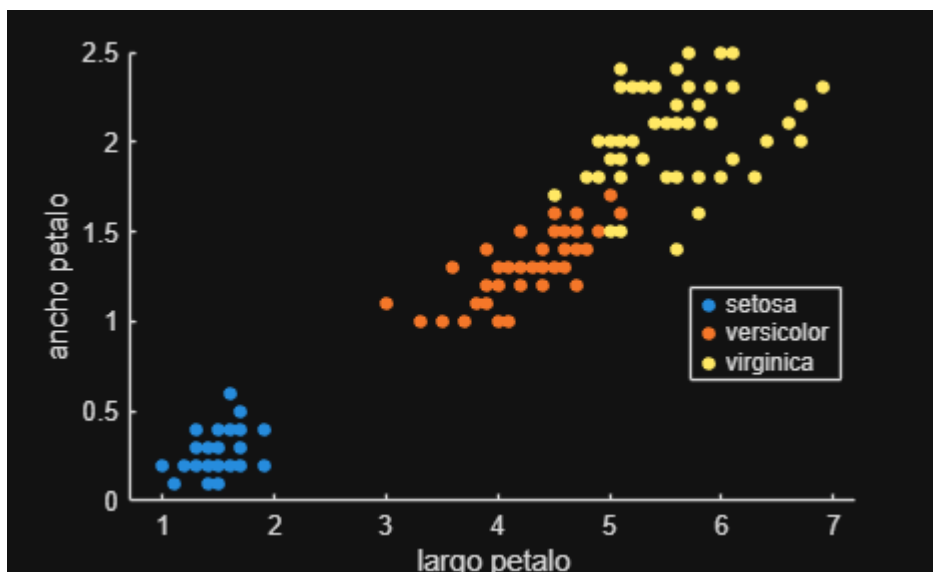
```
figure
boxplot(T[:,vars], 'Labels', vars); title('Iris crudo')
```



```
figure
histogram(T.clase); title('Balance de clases')
```



```
figure
gscatter(T.largo_petal, T.ancho_petal, T.clase)
xlabel('largo petalo'); ylabel('ancho petalo')
```



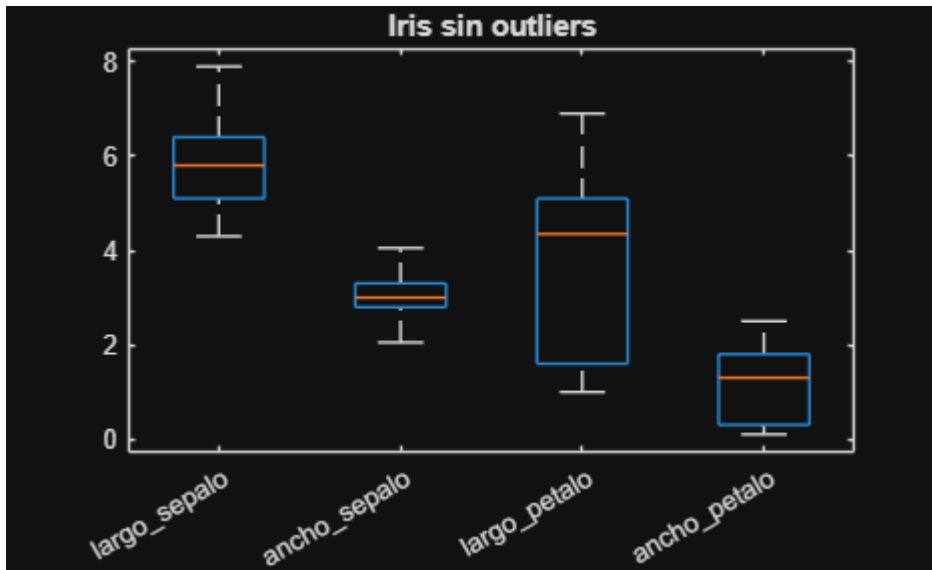
Paso 1: outliers con la regla 1.5*IQR, recortando al borde

```
T1_outliers = T0_crudo;
for k = 1:4
    x = T1_outliers.(vars{k});
    q = quantile(x, [0.25 0.75]);
    li = q(1) - 1.5*(q(2)-q(1)); ls = q(2) + 1.5*(q(2)-q(1));
    fuera(k) = sum(x < li | x > ls);
    x(x < li) = li; x(x > ls) = ls;
    T1_outliers.(vars{k}) = x;
end
```

```
fuera
```

```
fuera = 1×4  
0      4      0      0
```

```
figure  
boxplot(T1_outliers{:,vars}, 'Labels', vars); title('Iris sin outliers')
```



Paso 2: balanceo

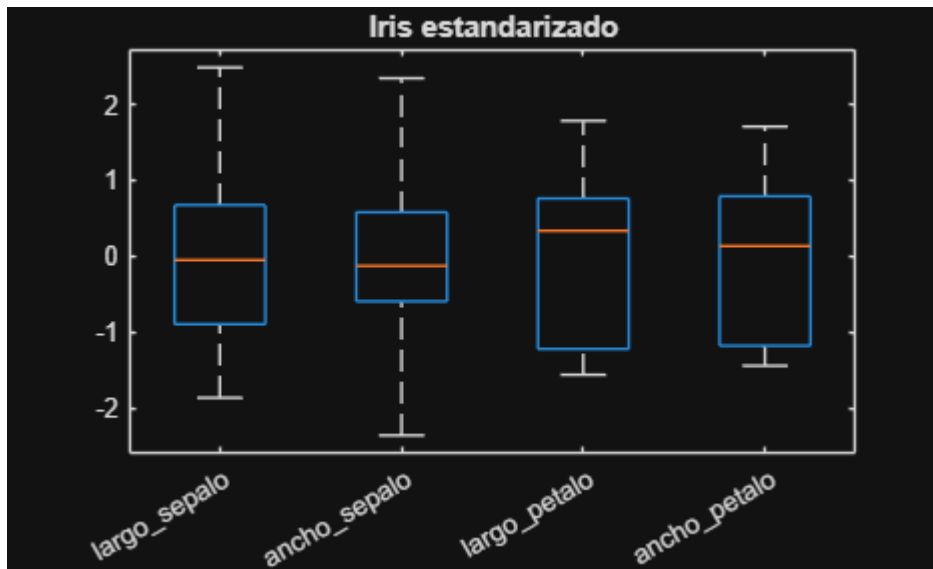
Iris ya viene con 50 de cada clase, no hay nada que hacer

```
T2_balanceado = T1_outliers;  
countcats(T2_balanceado.clase)'
```

```
ans = 1×3  
50    50    50
```

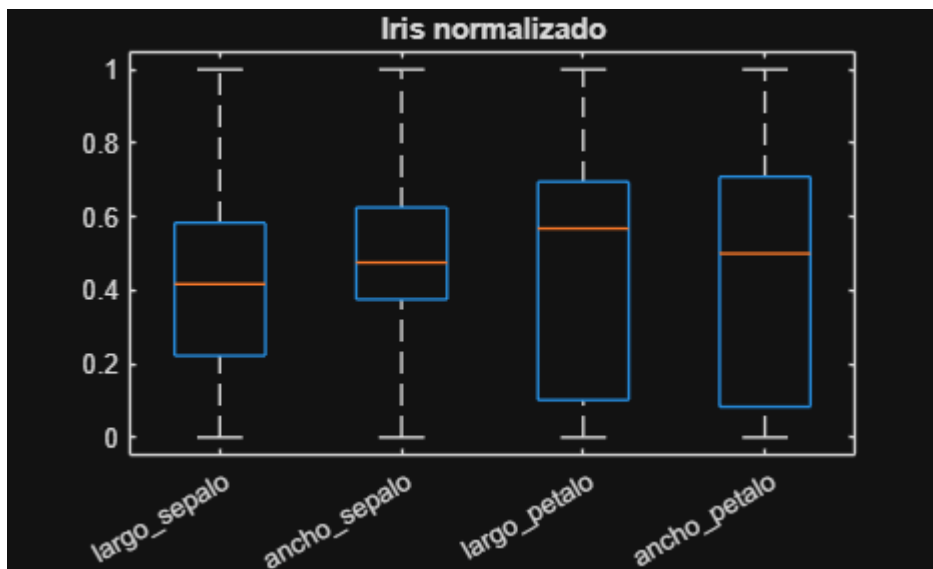
Paso 3: estandarizar

```
T3_estandarizado = T2_balanceado;  
T3_estandarizado{:,vars} = zscore(T2_balanceado{:,vars});  
figure  
boxplot(T3_estandarizado{:,vars}, 'Labels', vars); title('Iris estandarizado')
```



Paso 4: normalizar entre 0 y 1 (parte del balanceado, no del estandarizado)

```
T4_normalizado = T2_balanceado;
T4_normalizado[:,vars] = normalize(T2_balanceado[:,vars], 'range');
figure
boxplot(T4_normalizado[:,vars], 'Labels', vars); title('Iris normalizado')
```



```
save('iris_pasos.mat', 'T0_crudo', 'T1_outliers', 'T2_balanceado',
'T3_estandarizado', 'T4_normalizado')
```